

Delirium

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Introduction

Delirium is an acute, usually reversible, disturbance in mental status that develops over a short period of time (hours to a few days) and tends to fluctuate in severity. It results in a disturbance in cognition (e.g., memory deficit, disorientation, delusions, visual hallucinations, impaired language and visuospatial capabilities). The *Diagnostic and Statistical Manual of Mental Disorders*, 5th edition, text revision (DSM-5-TR) defines 3 subtypes of delirium:^[1]

- Hyperactive delirium
 - presents with increased psychomotor activity, possible mood lability or agitation, and/or refusal to cooperate with medical care
- Hypoactive delirium
 - presents with decreased psychomotor activity and possible sluggishness or lethargy that approaches stupor
- Mixed delirium
 - presents with a normal level of psychomotor activity and altered attention and awareness

While the pathophysiology of delirium is poorly understood, there are numerous known precipitating factors for its development. Although delirium may occur in any patient, those who are older, have sensory or pre-existing cognitive impairments, or are in postoperative, palliative or intensive care units are particularly predisposed. The more predisposing risk factors present, the fewer precipitating factors are needed to develop delirium (see [Table 2](#) and [Table 3](#)).

Delirium should be considered a medical emergency. When recognized in a community setting, patients require immediate evaluation, which is best and most efficiently completed in a hospital setting. Timely recognition of delirium can help trigger investigation into potentially life-threatening illnesses requiring urgent treatment. Delirium has been associated with increased mortality and duration of hospital admission;^[2] it may also contribute to long-term cognitive decline.^[3] This chapter focuses on the prevention, treatment and management of patients presenting with symptoms of delirium.

Goals of Therapy

- Prevent or minimize risk of delirium
- Prevent prolonged hospital stay through timely diagnosis and management of delirium
- Maintain safety of the patient, other patients and care providers through management of delirium-associated agitation

Investigations

- Timely recognition of the development of delirium is extremely important. A clinician or clinical team should become familiar and comfortable with the consistent use of the most appropriate screening tool for their clinical setting and use it regularly throughout a patient's stay. See [Table 1](#) for recommended and validated screening tools.
- If delirium is present, first consideration should be given to acute, life-threatening causes of delirium, such as hypoxia, hypotension, hypoglycemia, and drug intoxication or withdrawal. Then consider systematically identifying other potential causes.^[4] Note that multiple etiologies may be present. [Table 2](#) illustrates a mnemonic (I WATCH DEATH) for potential causes for delirium and can act as a guide for appropriate investigations. [Table 3](#) lists common medications associated with delirium.
- Obtain history from the patient and description of the patient's behaviour from staff and family members.
- Review medications and concomitant medical conditions.
- Conduct mental status examination, a complete physical examination and relevant laboratory investigations essential to the diagnosis.
 - Most patients will require the following laboratory investigations: complete blood count (including folate and vitamin B₁₂), electrolytes, metabolic panel, thyroid panel and urinalysis.
- Consider CT scan and/or EEG in patients with a suspected CNS pathology.^[4]

Table 1: Validated Screening Tools for Delirium

Screening Tool	Clinical Setting
Confusion Assessment Method (CAM)	High specificity and sensitivity for detecting delirium; appropriate for initial screening in emergency department. ^[5] CAM-ICU may be considered for patients in the ICU. ^[4]
4AT Method	Easy to use with good specificity and sensitivity for detecting delirium; can be completed in <3 min; appropriate for initial screening in emergency department. ^[6]
Single Question in Delirium	The question “Do you think [name of patient] has been more confused lately?” asked of a family member has good specificity and sensitivity in oncology and emergency room patients. ^{[7][8]}
Observational Scale of Level of Arousal (OSLA) ; Richmond Agitation and Sedation Scale (RASS)	Sensitivity and specificity of OSLA and RASS were >80% in older patients. ^[7]
Delirium Observation Screening (DOS) Scale	13-point screen for delirium in older patients, to be completed by a nurse and based on DSM-IV criteria for delirium. ^{[9][10]}
Delirium Rating Scale-Revised-98 (DRS-R-98)	16-item clinician-rated scale that can both diagnose and measure severity of delirium; useful for monitoring progress of patient. ^{[9][11]}
Memorial Delirium Assessment Scale (MDAS)	10-item clinician-rated scale used to quantify the severity of delirium; useful for monitoring progress of patient. ^{[9][12]}

Table 2: Conditions Commonly Associated with Delirium^{[4][13][14][15]}

Category ^[a]	Conditions
Infectious	One of the most common causes of delirium in older patients Encephalitis, meningitis, pneumonia, soft-tissue infections, syphilis, urinary tract infection
Withdrawal	From alcohol or benzodiazepines

Category ^[a]	Conditions
Acute metabolic	Acidosis, alkalosis, dehydration, electrolyte disturbances, hypercarbia, liver or kidney failure
Trauma	Heat stroke, burns, following surgery (prolonged surgeries may pose additional risk), pain
CNS pathology	Abscesses, hemorrhage, normal pressure hydrocephalus, seizure, stroke, tumor, vasculitis
Hypoxia	Anemia, carbon monoxide poisoning, hypotension, lung or heart failure, pulmonary embolism
Deficiencies	Sensory impairments; deficiencies of vitamin B ₁₂ , niacin, thiamine
Endocrinopathies	Hyper- or hypoglycemia, hyper- or hypoadrenocorticism, hyper- or hypothyroidism, hyper- or hypoparathyroidism
Acute vascular	Arrhythmias, hypertensive encephalopathy or shock, hypotension
Toxins or drugs	One of the most common causes of delirium in older patients General anesthesia, industrial poisons, medications (see Table 3), pesticides, solvents, illicit drugs
Heavy metals	Iron, bromium, lead, manganese, mercury

^[a] The listed categories spell out the mnemonic (I WATCH DEATH).

Table 3: Medications Associated with Delirium^{[14][16][17]}

Offending Agent	Comments
Alcohol	Alcohol withdrawal may also contribute to delirium (see Alcohol-Related Disorders).
Anticholinergics (e.g., atropine, benztropine, dicyclomine, oxybutynin,	Lower dose or discontinue, if possible.

Offending Agent	Comments
scopolamine)	
Antidepressants (e.g., paroxetine, TCAs)	Consider replacing with SSRIs (other than paroxetine) or SNRIs.
Antiemetics (dimenhydrinate, prochlorperazine, promethazine)	Consider replacing with domperidone, metoclopramide or 5HT3 receptor antagonists (e.g., ondansetron).
Antihistamines (e.g., brompheniramine, chlorpheniramine, diphenhydramine, doxylamine)	Replace with second-generation non-sedating antihistamines.
Antipsychotics (especially low-potency FGAs)	Lower dose or discontinue, if possible (see Deprescribing). If necessary for agitation, use low doses of high-potency agents (see Table 4).
Barbiturates	Replace with alternative or discontinue, if possible (low seizure risk, no recent history of seizures).
Benzodiazepines and other hypnotics (e.g., zolpidem, zopiclone)	Replace with sleep hygiene measures. If patient is a chronic user, do not suddenly discontinue, as withdrawal can contribute to delirium. Consider lowering dose (see Deprescribing).
Corticosteroids	Use lowest effective dose for the shortest duration of time.
H ₂ -receptor blockers (e.g., cimetidine, famotidine, ranitidine)	Replace with antacids, PPIs.
Muscle relaxants (e.g., cyclobenzaprine, orphenadrine)	Lower dose or discontinue, if possible.
Opioids (particularly meperidine)	Replace with acetaminophen or NSAIDs, if possible. Note, uncontrolled pain can contribute to delirium; risks and benefits of opioids should be considered depending on pain severity.

Abbreviations: FGA = first-generation antipsychotic; SNRI = serotonin-norepinephrine reuptake inhibitor; SSRI = selective serotonin reuptake inhibitor; NSAID = nonsteroidal anti-inflammatory drug; PPI = proton pump inhibitor; TCA = tricyclic antidepressant

Prevention

Strong evidence supports multicomponent interventions for the prevention of delirium in hospitalized patients, which include monitoring and correction of electrolytes/glucose, hydration, regulation of bowel and bladder function, pain relief, avoidance of polypharmacy, and oxygen therapy.^{[17][18]} There is also evidence that monitoring depth of anesthesia can reduce occurrence of delirium after general anesthesia.^[4]

The Hospital Elder Life Program (HELP) is effective for reducing incidence of delirium and rate of falls in older patients.^[19] HELP includes screening older patients upon hospital admission for risk factors of delirium and targeted management of these risks by an interdisciplinary team. This team may include a geriatrician, a geriatric nurse therapist, an elder-life specialist and trained volunteers.^[20]

There is insufficient evidence to support the general use of pharmacologic agents, such as antipsychotics, for the prevention of delirium.^[17] Olanzapine, risperidone, ketamine, melatonin and ramelteon (not available in Canada) may have preventative effects, but further study is required.^{[21][22][23]}

Dexmedetomidine, a sedative typically used in the ICU, may be considered as a preventative measure for patients at high risk of delirium who also require sedation. Evidence is conflicting, and further studies are required in order to generalize its efficacy in the prevention of delirium.^{[21][24][25][26]}

Therapeutic Choices

Nonpharmacologic Choices

First, address potential causes of delirium identified on patient assessment. If symptoms persist, then patients with delirium should be managed with nonpharmacologic and behavioural methods to prevent and reduce agitation. These include frequent reorientation with the use of clocks, calendars, and pictures of family members as well as reminders by staff of the date, place and circumstances of the patient's stay in hospital. Patients should be mobilized as early as possible, and sleep patterns should be maintained as much as possible. It is important to ensure that patients who require glasses or hearing aids have them available.

Communication of diagnosis to patient and caregivers, and encouragement of involvement of caregivers, may be helpful. Providing ongoing engagement and support with an aim to prevent complications of delirium such as falls, immobility, dehydration and isolation may also be of benefit.^[4]

Pharmacologic Choices

Pharmacologic options discussed in this chapter are targeted toward management of agitated behaviours and not treatment of delirium. Minimizing or removing drugs that may be contributing to the development

of delirium should be the first step in pharmacological management (see [Table 3](#)).

If there are ongoing concerns of risk of harm to patient, staff or other patients posed by the agitated, delirious patient and nonpharmacologic options have been exhausted, it may be necessary to use pharmacologic options. While off-label uses of antipsychotic medications are the primary treatment options (see [Table 4](#)), it should be noted that systematic reviews have shown limited benefit in reduction of severity or duration of delirium with these medications.^[27] They have also been found to have no impact on length of stay in admission or associated morbidity/mortality. The goal of therapy is to ensure safety of the patient, other patients and staff by minimizing agitation. Use antipsychotics with caution, at low doses, and for as short a duration as possible, particularly in older patients and those with dementia who are at an increased risk of antipsychotic-associated stroke and death.^{[28][29]} Small, regular doses are preferred over PRN dosing. Taper and discontinue antipsychotics upon resolution of delirium (see [Deprescribing](#)).

Haloperidol is a first-generation antipsychotic (FGA) that is the most studied medication for management of agitated delirium. When used parenterally, extrapyramidal symptoms (e.g., akathisia, dystonia, parkinsonism) can be common adverse effects. **Loxapine**, a mid-potency FGA, and second-generation antipsychotics (SGAs) such as **olanzapine**, **risperidone** and **quetiapine** are also commonly used to treat agitation related to delirium.^{[30][31][32]} SGAs may be safer than haloperidol while being equally effective in the management of agitation.^{[33][34]} Oral disintegrating tablets (e.g., olanzapine) may be beneficial for patients who are not willing to swallow tablets.^[35] Parenteral administration (e.g., haloperidol, loxapine, olanzapine) may be required for highly agitated or uncooperative patients who cannot or are unwilling to take oral medications. Loxapine and olanzapine have significant anticholinergic properties and may exacerbate cognitive impairment; they are not recommended for use in older patients.^[16] Quetiapine also has significant anticholinergic properties, but to a lesser extent than loxapine and olanzapine.

Reserve **benzodiazepines** (preferably those with a longer half-life such as **diazepam** or **chlordiazepoxide**) for cases of delirium where the symptoms are attributed to alcohol or benzodiazepine withdrawal (see [Alcohol-Related Disorders](#)). For all other etiologies of delirium, benzodiazepines can contribute to the disease process and should be avoided.^{[14][36][37][38]} Although a network meta-analysis demonstrated that the combination of **haloperidol** and **lorazepam** might be effective for the management of delirium (not-related to alcohol or benzodiazepine withdrawal),^[21] further studies are required before this combination can be recommended, as the results differ from most other data.

Although there is growing evidence to support **dexmedetomidine** as a treatment option as it may shorten the duration of delirium,^[39] the evidence is not sufficient to recommend for routine use. There may be an increased mortality risk in patients <65 years of age with a nonoperative diagnosis.^[40]

The use of **ketamine** for the management of delirium-associated agitation is under investigation; there is currently insufficient data on this topic to make any recommendations.^[41]

Choices during Pregnancy and Breastfeeding

Employ nonpharmacologic methods as first-line treatment in any patient presenting with delirium, especially those who are pregnant or breastfeeding. A discussion of general principles on the use of medications in these special populations can be found in [Drugs Use during Pregnancy](#) and [Drug Use during Breastfeeding](#). Other specialized reference sources are also provided in these appendices.

Antipsychotics in Pregnancy

There are *no studies* assessing the use of medications for management of symptoms of delirium in pregnant patients. Use pharmacologic treatments for delirium only in situations where there is significant risk of harm to the patient, infant, staff or other patients on the treatment unit, and with informed consent. Teratogenic effects, including major congenital malformations, have not been demonstrated with the use of antipsychotics during pregnancy for the treatment of psychoses.^[42] Hypertonia in neonates has been noted following prepartum use of high-potency FGAs (e.g., haloperidol). For more information on the use of antipsychotics in pregnancy and breastfeeding, see [Schizophrenia and Related Psychotic Disorders](#).

Antipsychotics in Breastfeeding

Antipsychotics are generally passed into breast milk and are detectable at levels considered compatible with breastfeeding; however, data in this patient population is limited and long-term studies are needed. If antipsychotics are used, monitor the infant for adverse effects such as lethargy, drowsiness and developmental milestones (long-term use). Haloperidol at doses of 10 mg/day or less has not displayed any effect on the infant, based on limited studies. Higher doses are not recommended.^[43] Loxapine is not a preferred agent in those who are breastfeeding due to a lack of available data. Of the SGAs, olanzapine is the preferred agent in breastfeeding patients based on over 100 documented cases of low infant exposure (low relative infant dose in breast milk).^[44] Quetiapine also demonstrated low infant exposure, although the number of documented cases was less than olanzapine. Risperidone was associated with more infant exposure than other SGAs and may not be a preferred option.

Therapeutic Tips

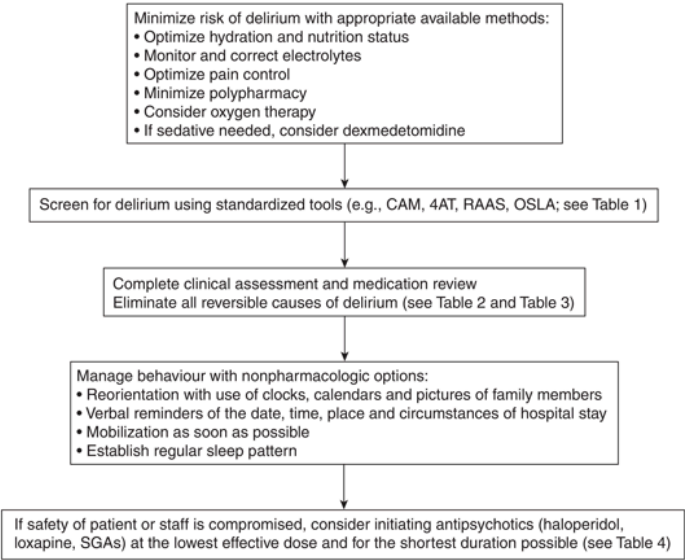
- There may be pre-existing cognitive impairments that have been undetected or exacerbated with delirium. Consideration should be given to cognitive and functional assessments.
- Reserve pharmacological interventions for those patients presenting with agitation that endangers themselves, other patients or staff. Re-evaluate the need for continued therapy regularly. Taper and discontinue antipsychotics as soon as delirium resolves. Ensure discontinuation prior to discharge.
- Use lower doses of antipsychotics in older patients.
- Antipsychotics can cause QTc prolongation; carefully consider the potential risks and benefits before using in patients at increased risk (female,* older patients, bradycardia, pre-existing heart disease,

hypokalemia, hypomagnesemia).^{[45][46]} (*Note: Studies presenting this finding classify patients as male or female. CPhA is aware these terms may not reflect the identity of the individual patient presenting and are not inclusive of all patients. Each patient should be considered individually.)

- Regular assessments (daily or more frequently) via screening tools are useful to recognize fluctuation, improvement or worsening of delirium.^{[4][11]} See [Investigations](#) for commonly used screening tools.

Algorithms

Figure 1: Prevention and Management of Delirium



Abbreviations

CAM = Confusion Assessment Method; RAAS = Richmond Agitation and Sedation Scale; OSLA = Observational Scale of Level of Arousal; SGA = second-generation antipsychotic

Drug Table

Table 4: Drugs Used for the Management of Agitated Delirium^{[4][14]}

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects	Drug Interactions	Comments
Drug Class: Antipsychotics, first-generation				
haloperidol generics PO: <\$2 IM: \$2–4	0.5–1 mg Q6–12H PRN PO/IM; maximum 2 mg/day Geriatric: 0.25–0.5 mg; maximum 2 mg/day	Sedation, parkinsonism, akathisia, acute dystonia, neuroleptic malignant syndrome. Haloperidol: QT _c prolongation.	Additive effects with other CNS depressants, antagonism of dopamine agonists. Haloperidol: avoid combining with other drugs that increase QT _c interval.	Incidence of extrapyramidal symptoms is dose-related.
loxapine Xylac <\$2	2.5–10 mg BID PRN PO/IM; maximum 20 mg/day	Sedation, parkinsonism, akathisia, acute dystonia, neuroleptic malignant syndrome. Anticholinergic effects.	Additive effects with other CNS depressants, antagonism of dopamine agonists. Haloperidol: avoid combining with other drugs that increase QT _c interval.	Not recommended in the older patients because of strong anticholinergic properties.
Drug Class: Antipsychotics, second-generation				
olanzapine, oral (including oral disintegrating tablets) Zyprexa, Zyprexa Zydis, generics <\$2	2.5–10 mg PO BID PRN; maximum 20 mg/day	Anticholinergic effects, akathisia/EPS, dizziness, neuroleptic malignant syndrome. Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) is a rare	Additive sedation with CNS depressants; antagonism of dopamine agonists; may potentiate antihypertensive drug effects.	Not recommended in the older patients because of strong anticholinergic properties.

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects	Drug Interactions	Comments
		but very serious adverse effect. (see Drug-Induced Skin Reactions).		
olanzapine, intramuscular Zyprexa Intramuscular \$37/10 mg vial	2.5–10 mg IM; may repeat in 2 h and 6 h PRN to a maximum of 30 mg/24 h Use maximum of 2.5 mg/dose in debilitated patients	Anticholinergic effects, akathisia/EPS, dizziness, neuroleptic malignant syndrome. Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) is a rare but very serious adverse effect. (see Drug-Induced Skin Reactions).	Additive sedation with CNS depressants; antagonism of dopamine agonists; may potentiate antihypertensive drug effects. Should <i>not</i> be administered simultaneously with parenteral benzodiazepines due to reports of cardiac and respiratory problems including death.	Not recommended in the older patients because of strong anticholinergic properties.
quetiapine Seroquel, generics <\$2	12.5–50 mg PO BID PRN; maximum 100 mg/day Geriatric: 12.5–25 mg; maximum 50 mg/day	Anticholinergic effects, akathisia/EPS, dizziness, neuroleptic malignant syndrome. Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) is a rare but very serious adverse effect. (see Drug-	Additive sedation with CNS depressants; antagonism of dopamine agonists; may potentiate antihypertensive drug effects.	

Drug/Cost ^[a]	Dosage ^[b]	Adverse Effects	Drug Interactions	Comments
		Induced Skin Reactions).		
risperidone  Risperdal , Apo-Risperidone, other generics <\$2	0.5–1 mg PO BID PRN; maximum 2 mg/day Geriatric: 0.25–0.5 mg; maximum 2 mg/day	Anticholinergic effects, akathisia/EPS, dizziness, neuroleptic malignant syndrome. Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) is a rare but very serious adverse effect. (see Drug-Induced Skin Reactions).	Additive sedation with CNS depressants; antagonism of dopamine agonists; may potentiate antihypertensive drug effects.	

^[a] Cost of 1-day supply unless otherwise specified; includes drug cost only.

^[b] Dosages are based on the clinical experience of the authors. Dosages may vary based on institution-specific delirium protocols.

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).

Abbreviations:

CNS = central nervous system; EPS = extrapyramidal symptoms

Suggested Readings

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